

Hotel Booking Analysis

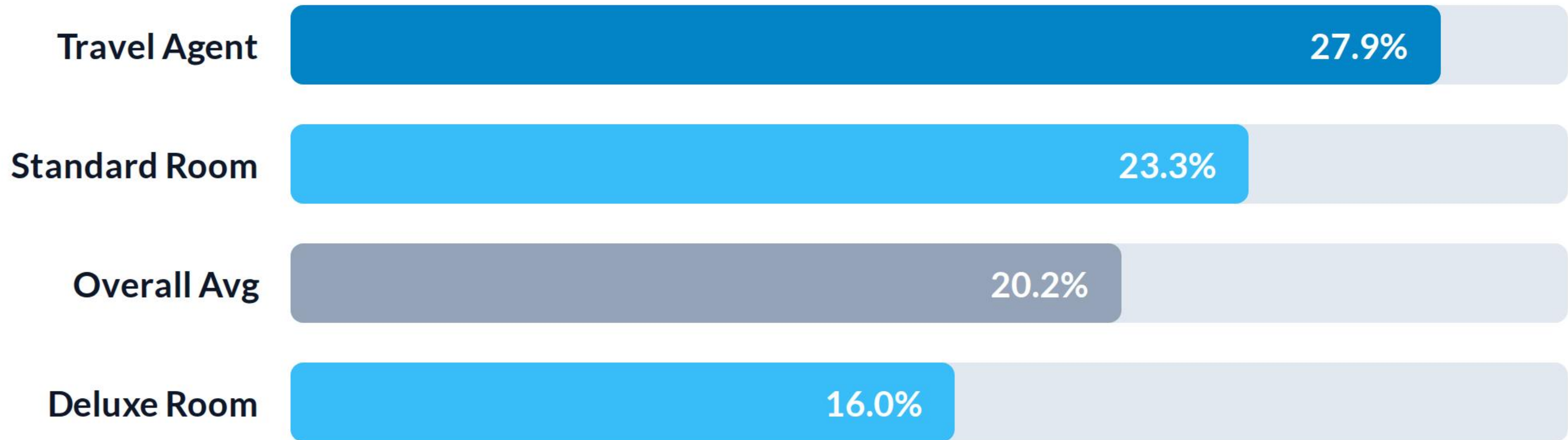
Date: June 13, 2026

Dataset Overview

- 30,000 booking transactions recorded.
- Timeframe spans from **April 2024 to April 2025**.
- 24 columns covering vital metrics: bookings, revenue, and cancellations.
- 3 booking channels: **Web, Mobile App, Travel Agent**.
- 3 primary room types: **Standard, Deluxe, Suite**.
- Overall cancellation rate stands at **20.23%**.
- Overall failed bookings account for **7.53%**.



Key Observation 1: Cancellations



- *Travel Agents* have the highest cancellation rate at 27.9%.
- *Standard rooms* cancel significantly more frequently (23.3%) compared to *Deluxe rooms* (16.0%).
- *Star rating* showed **no significant impact** on cancellations (rates range tightly between 19.8% to 21.3%).

Key Observation 2: Channel & Revenue



- The **Web channel** generates the highest average booking value (₹28,191).
- The **Mobile App** lags behind noticeably in transaction value (₹21,351).
- **Travel Agents** sit in the middle for revenue but represent the worst cancellation rates overall.

Key Observation 3: Seasonality



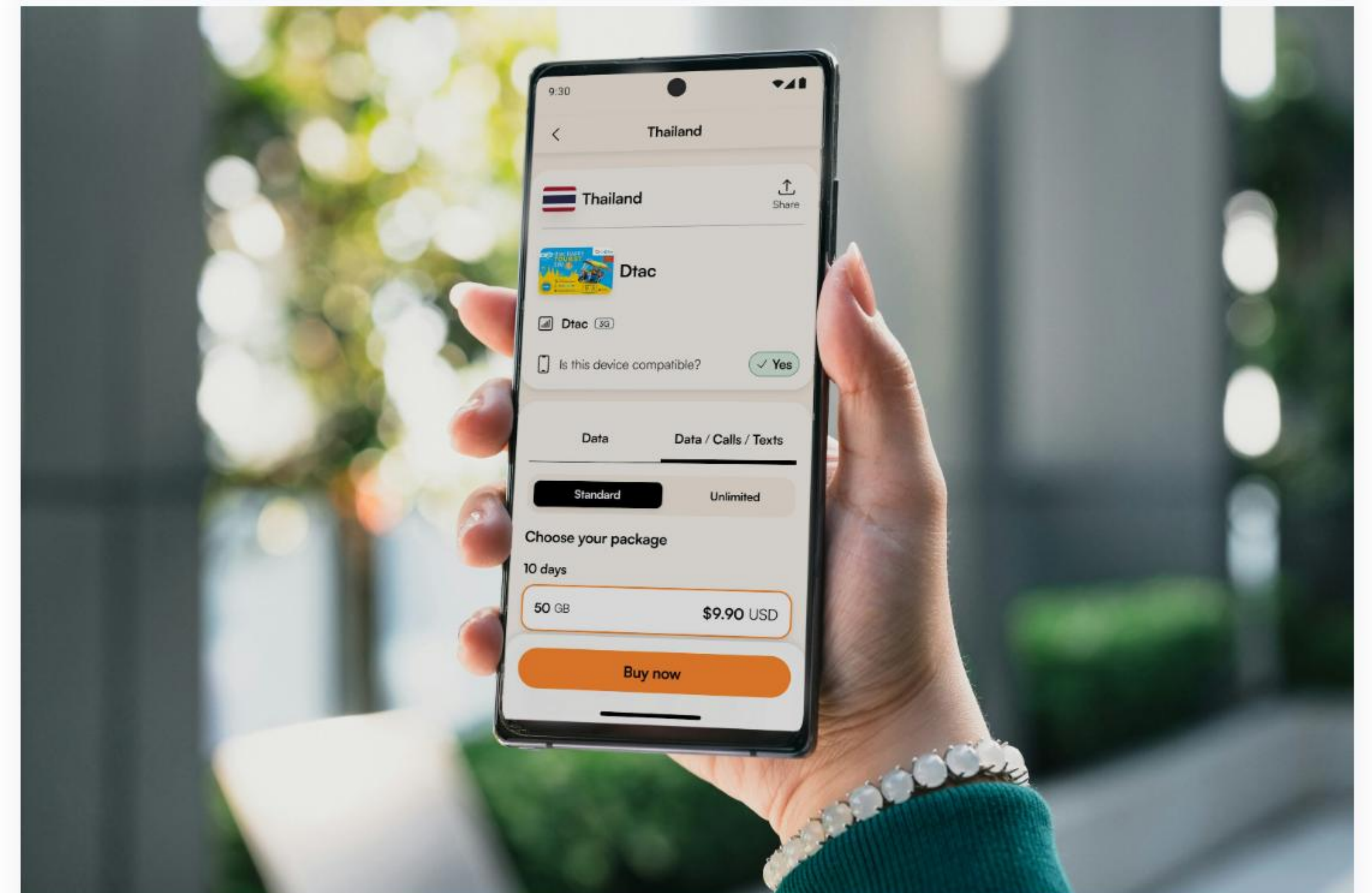
- *Peak travel months historically hit heavily in **July and August**.*
- ***January** represents the weakest month, securing only 339 recorded bookings.*
- ***Leisure travelers** form the dominant demographic, making up 60.4% of all bookings.*

Root Cause Analysis

Observation	Root Cause
High Travel Agent Cancellations	Agents typically book on behalf of customers, resulting in less personal financial commitment upfront.
High Standard Room Cancellations	Lower price point equates to an easier cancellation decision, with less financial loss felt by the customer.
July-August Peak Volume	Direct alignment with summer holidays, school breaks, and the universally peak leisure travel season.
Web Channel Drives Highest Value	Web users generally plan more extensively and book longer stays, indicating higher-intent purchases.

Recommendation 1: Reduce Cancellations

- 👉 **Non-Refundable Incentives:** Introduce non-refundable rates tailored for Standard rooms, offering a 5-10% discount.
- 💰 **Cancellation Fees:** Begin charging a small cancellation fee strictly on Travel Agent bookings.
- 📅 **Flexible Modifiers:** Offer a free date-change option to minimize outright booking cancellations.
- 🔔 **Automated Alerts:** Send an automated reminder directly to the customer 7 days before check-in.



Recommendation 2: Profitability & Repeat Bookings

- 🎯 **Prioritize Web Channel:** Shift marketing spend to heavily prioritize the Web channel (highest value + lowest cancellation = best overall ROI).
- 📈 **Strategic Upselling:** Create automated funnels to upsell Standard room bookers to Deluxe rooms right at the checkout stage.
- 💎 **Loyalty Programs:** Launch a robust loyalty program explicitly targeting Leisure travelers (60% of bookings, highest repeat potential).
- 💰 **Cashback Incentives:** Replace traditional upfront coupon discounts with cashback rewards strategically designed to lock in repeat customers.

Recommendation 3: Pricing & Channel Strategy

-  **Dynamic Pricing:** Apply aggressive dynamic pricing models during the high-demand peak months of July and August.
-  **Early Bird Promotions:** Run targeted early bird campaigns in historically slow months (Jan, Nov, Mar) to stimulate volume ahead of time.
-  **Reduce Agent Dependency:** Scale back reliance on Travel Agents to drastically cut both high commission costs and cancellation rates.
-  **Mobile App UX:** Invest significantly in Mobile App UX improvements to close the current ₹6,840 average booking value gap with the Web.

Summary & Key Findings



Cancellations

Travel Agents and Standard rooms are the primary, outsized drivers of booking cancellations.



Channel Value

The Web channel stands out decisively as the most valuable and highly reliable revenue stream.

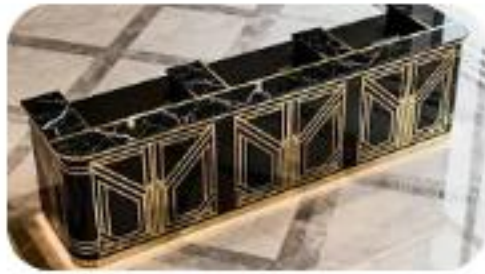


Seasonality

Summer months dominate booking volume, emphasizing the need for targeted promotions in low months.

Thank You!

Image Sources



<https://dzsculpture.com/wp-content/uploads/2025/11/7-9.webp>

Source: www.dzsculpture.com



<https://images.unsplash.com/photo-1761293877325-912e741f1a96?fm=jpg&q=60&w=3000&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D>

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